Effect of pH and temperature on the microbiome of developing fish

Clara Tucker[1], Megan Hahn[1], Janet Nye[1], Hannes Baumann[2], Nolwenn Dheilly[1]

[1]School of Marine and Atmospheric Sciences, Stony Brook University, Stony Brook, NY 11794-5000, USA

[2]Department of Marine Sciences, University of Connecticut, 10801 Shennecossett Road, Groton, CT 06340, USA

SoMAS School of Marine and Atmospheric Sciences
STONY BROOK UNIVERSITY

Introduction

- Global climate change causes a rise in ocean temperature and acidity, which may affect the health of marine animals.
- The composition of the microbiome of a developing organism can affect the development of its immune system, and thus impair its health for the rest of its life.
- The microbiome of fish is strongly dependent on environmental conditions.
- In this study, we determine how temperature and pH can independently and through interaction affect the microbiome of developing Atlantic silversides (*Menidia menidia*).

Figure 1. Atlantic silversides (*Menidia menidia*) are very abundant along the coast of New-York and Connecticut

Methods

Experiment 1

10 days post hatching, n=3 per treatment		
	17˚C	24˚C
pH = 8.2 (Control)	17C	24C
pH = 7.5 (High CO_2)	17H	24H

1 days post hatching, n=3 per treatment		
	17˚C	24˚C
pH = 8.2 (Control)	17C	24C
pH = 7.5 (High CO_2)	17H	24H

Experiment 2

10 days post hatching, n=6 per treatment			
	17˚C	20˚C	24˚C
pH = 8.2 (Control)	17C	20C	24C
pH = 7.5 (High CO_2)	17H	20H	24H
pH = 7.2 (Very High CO_2)	17VH	20VH	24VH

- Silversides were conditioned in Spring of 2016 using complete cross experimental designs.
- The microbiome composition and diversity was determined through sequencing of the bacterial 16S rRNA gene V4 hypervariable region.
- The bacterial load was estimated by RT-qPCR with Universal 16S rRNA bacterial primers.

Results

- Older larval fish have higher bacterial diversity than younger larval fish ($p < 0.05$). Increased temperature but not pH affects bacterial diversity ($p < 0.05$)

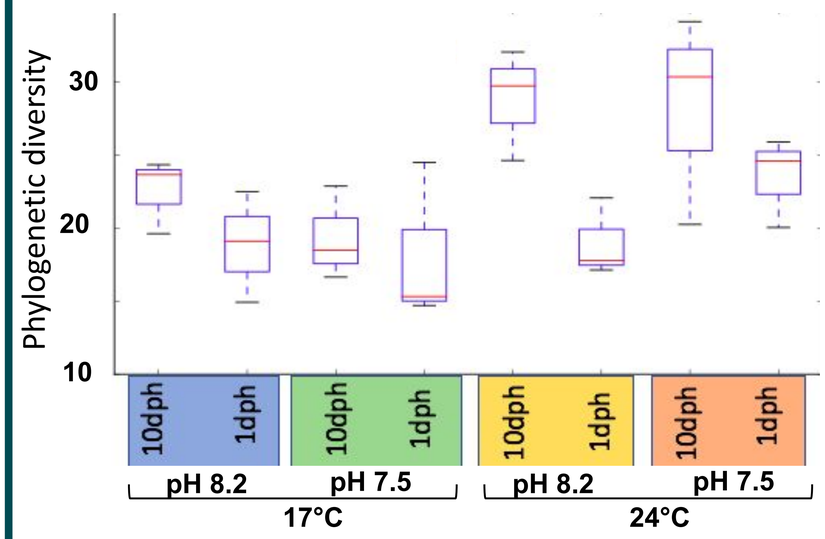

Figure 2. Abundance weighted Phylogenetic measure of Alpha diversity in silversides collected 1 day post hatching (1dph) or 10 days post hatching (10dph) and conditioned under different temperatures and pH.

- *Vibrio sp.* are among the most abundant bacteria in developing silversides, and their abundance vary significantly in response to variations in pH and temperature (DESeq, $p<0.05$).
- Known pathogenic species of *Vibrio* are more abundant under predicted future environmental conditions.

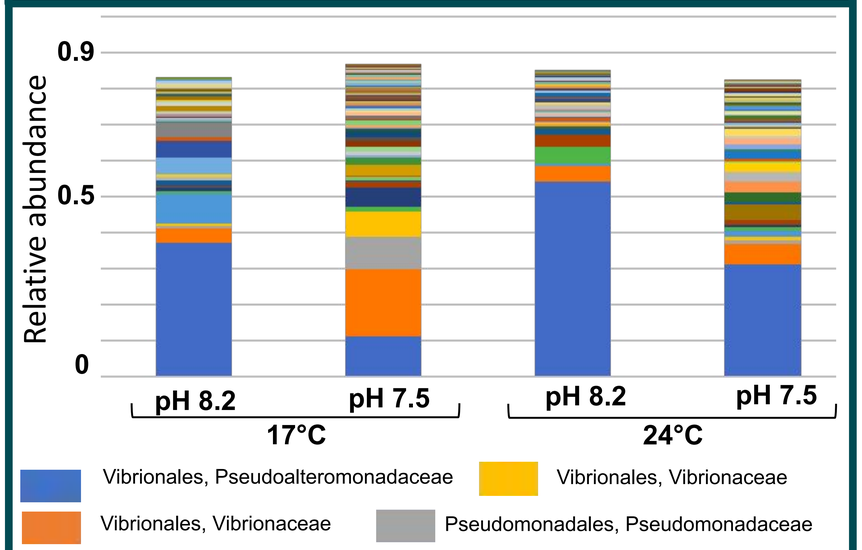

Figure 3. Bacteria representing more than 1% total abundance in 10 dph silversides conditioned under different tempoeratures and pH. The order and family of the four most abundant bacteria is provided.

- Results suggest that lower pH is associated with lower bacterial load, except at 24°C when bacterial load is high under all pH conditions.

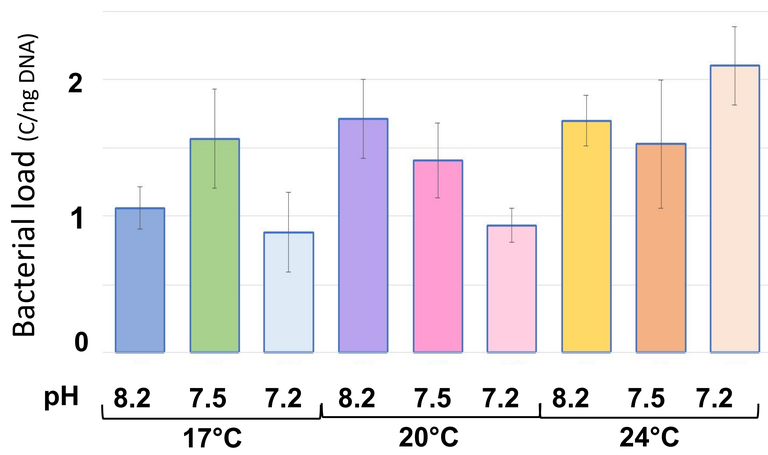

Figure 4. Bacterial load expressed as the number of bacterial cells per ng of extracted DNA measured in silversides collected 10dph under different tempoerature and pH.

On-going and future analyses

- Extend the study of the microbiome diversity of 10 dph larvae to combine results from Experiments 1 and 2.
- Review the literature for the effects of environmental stressors on related species of marine fish.
- Measure the expression of immune response genes in developing larvae collected from the same experiments.

Discussion

- The microbiome of fish hatched under different environmental conditions does not differ dramatically.
- On-going analyses of the microbiome of silversides collected 10 days post hatching indicate that **temperature and pH have an interacting effect** on bacterial load, and microbiome composition.
- Temperature and pH can impact the microbiome of developing fish, suggesting that environmental changes in coastal and estuarine body of waters, often called nurseries of the sea, could have long-term effects on fish health and fisheries.

Acknowledgments: This work was supported by NSF award #1536165 to HB, and Stony Brook University starting funds to NMD and JN.

Stony Brook University School of Marine and Atmospheric Sciences